TINA — a distributed approach to service management

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The Telecommunications Information Networking Architecture Consortium (TINA-C) was formed 5 years ago to develop an architecture to address the radical business and technological changes then envisaged in the future telecommunications world. Central to the proposed solution is an infrastructure known as the TINA distributed processing environment (DPE). The TINA service architecture, which defines a set of concepts and rules for the management of a wide range of services, is of particular interest here. This paper outlines the current status and output of the TINA Consortium and goes on to describe a collaborative implementation project which uses many of the TINA concepts and specifications to build a pan-European service platform. An important vision within TINA is that of open interfaces and interoperability in a heterogeneous environment. The extent to which this goal has been achieved is discussed in the light of experience.

1. Introduction

TINA-C [1] was set up in 1993 to address a number of requirements and opportunities relating to rapid changes in regulation, business markets and technology. The consortium was global in nature and had over 50 members including telecommunications network providers, telecommunications and information technology suppliers, and research organisations.

The original ideals are still extremely relevant today. From a customer viewpoint these include:

- customers of all types demanding a wide range of customised services, from any location, at any time,
- customers demanding new services very quickly after concept demonstration,
- services supplied at low cost.

From a service supplier and manufacturer's viewpoint these include:

- the development of a generic service architecture which allows interoperability on a global scale and reduces time-to-market,
- the provision of services on heterogeneous network architectures and topologies.

The consortium adopted the slogan: 'A co-operative solution in a competitive world'. By the end of 1997, TINA-C had defined a coherent set of concepts that viewed the problem space from a number of angles. A number of important specifications were produced, including a business model, service architecture, network resource architecture and DPE architecture. The consortium has clearly been successful at influencing thinking in the telecommunications world. It has worked, and continues to work, closely with standards bodies such as the Object Management Group (OMG), the TeleManagement Forum (formerly NMF), the ATM Forum and the ITU-T in a number of areas.

In addition to the architectural work, a number of TINA auxiliary projects have publicly demonstrated a wide range of the technologies and concepts. However, it has been recognised that, while the current state of TINA allowed for the development of advanced prototypes, moving to commercial deployment requires stable specifications to stimulate the market for compliant products. In order to resolve specifications for particular inter-business interactions and facilitate the market-driven adoption of the architecture, the consortium is continuing for a further two years from 1998. The revised consortium is driven by Working Groups each addressing just one key area (e.g. migration of existing IN architectures to TINA principles, applying the TINA architecture to mobile terminals and users, and definition of a full set of interfaces to support the service architecture). The Working Groups issue requests for proposals (RFPs) on specific topics. Stable and manageable specifications, standardised as necessary, will increase product availability and will lead to real opportunities for both operators and vendors.

This paper reviews the current status of important parts of the TINA architecture. The main omission is the network resource architecture. This is generally regarded as being less mature than other parts of the architecture, and as yet has a number of unresolved issues. More details of the TINA network resource architecture may be found in Loosemore et al [2].

An implementation project is then discussed in which a pan-European service platform has been built and tested. In this work the latest specifications from TINA Working Groups have been used and various key TINA requirements have been tested including the ease of interworking between different commercial distributed processing environments (DPEs) and the practicality of the specifications.

2. The TINA business model

A key principle of TINA is that it should be an open architecture, capable of being used in a multiple stakeholder (user/service provider/network operator) environment. The business model is intended to provide a framework to ensure that this will be the case. The business model first defines a common business framework that allows business roles and reference points to be created and modified within TINA. Reference points specify the interactions between business roles and form the basis for compliance with TINA.

In addition, the business model defines 'an initial set' of business roles and relationships (reference points) which are likely to be useful in the short term. These roles are:

- consumer — makes use of services,
- retailer — serves consumers,
- broker — provides information to allow location of stakeholders and services,
- third party service provider — supports retailers or other third party service providers with services,
- connectivity provider — manages a network to support user connections or DPE connectivity.

These roles are shown in Fig 1, along with the initial set of relationships between them. It is certainly possible to take exception with the initial roles and relationships. These were identified with a number of implicit assumptions. However, other roles and relationships can be added within the common business framework to reflect the requirements of the market being considered. The business modelling concepts are more generally applicable and are being regarded with some interest in a number of standards bodies.

A stakeholder that plays a business role and conforms to the relevant reference points should be able to participate fully in a TINA system. It is also possible for a single enterprise to play more than one role at any time. For example, BT could be a retailer, a third party service provider and a connectivity provider in the delivery of some services. Again, as long as the interactions with roles outside the enterprise conform to TINA reference points, the enterprise is TINA-compliant.

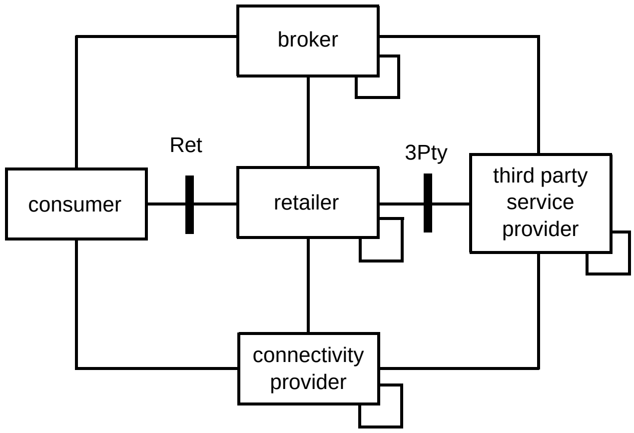

Fig 1 The TINA business model.

An example of a simple usage scenario, in which a different enterprise takes each role, might be:

- a consumer requests a service from a retailer,
- the retailer obtains service components from a third party service provider,
- the retailer obtains network connectivity from a connectivity provider,
- the consumer receives service.

As far as the consumer is concerned, the retailer provides the service (and a single bill). In addition, the third party service provider and connectivity provider deal only with the retailer and do not have to manage relationships with many customers. This contrasts with the typical Internet model where there is direct interaction between consumer and content provider.

The important reference points for this work are labelled in Fig 1. These are 'Ret' (between consumer and retailer) and '3Pty' (between retailer and third party service provider). A reference point consists of a set of interface specifications in OMG interface definition language (IDL) and informal descriptions of the domain interactions including clarifications of the intended use of the interfaces. 'Ret' is largely stable now. '3Pty' has not yet been specified, although the assumption is that much of 'Ret' should be reusable.

3. Distributed processing environment

The DPE is the basic infrastructure underlying any TINA architecture. It is ubiquitous in any TINA system and provides the basis for object communication. The DPE architecture essentially defines a set of requirements on a platform technology. OMG's CORBA (Common Object Request Broker Architecture [3]) is the dominant technology for realising a TINA DPE in existing prototypes, and is generally sufficient. However, there are a number of outstanding issues not yet satisfied by CORBA, including multiple interfaces, stream interfaces and support for mobility.

It is also important that CORBA products should meet the stringent non-functional requirements (scalability, fault tolerance, etc) of typical telecommunications systems.

Activity of TINA member companies within the OMG should ensure that the requirements of the telecommunications sector are fully supported as CORBA evolves.

As the DPE architecture defines its characteristics so that they are independent of any particular platform technology, it is possible that alternatives to CORBA could be considered in the future.

4. Service architecture

The TINA service architecture [4] is aimed at the construction, deployment, operation and withdrawal of TINA services. An important feature is the separation of access, usage and communication aspects of the architecture. The access part deals with a user's universal access to services. The usage part allows use of a particular service, which may also involve the communication part for control of connections. For example, in a multiparty videoconferencing service, an audio/video (AV) stream path between all parties needs to be established. Set-up and basic stream control is handled in the communication part, separate from the main service logic. This separation allows uniform access to a wide range of services and also allows new services to be introduced, independent of the underlying network. Personal mobility and session mobility are also supported.

The service architecture also identifies a computational model which includes service management components, such as service life cycle, accounting, and subscription management. Applications can be rapidly developed by making use of these generic components.

The service architecture is recognised as a particularly strong part of TINA and the basic approach is now being used in service platforms which are appearing on the market. However, some reference point specifications are not yet sufficiently mature for TINA compliance to be considered as the basis for interoperability between different vendors.

5. TINA-compliant pan-European platform

BT is currently involved in a European collaborative project (EURESCOM P715, 'EURESCOM services platform' [5]), with five other partners — Deutsche Telekom AG (DT), France Telecom (FT), FINNET Group (AF), KPN-Royal PTT Netherlands NV (NL) and Telecom Eireann (TI). The primary goal is to investigate the delivery of telecommunications services in a multiple-provider environment, based on open interfaces. The approach has been to use TINA specifications where appropriate and available. To make the results of the project practical and meaningful, independent development and a heterogeneous infrastructure have been encouraged between the six partners involved. The DPE used is a pan-European platform based on a variety of CORBA 2.0 Object Request Brokers (ORBs), communicating via the Internet inter-orb protocol (IIOP) over a dedicated TCP/IP network over N-ISDN. This choice of network was made as a convenient way of achieving the required connectivity with a low-latency variance network. The public Internet was often found to give unacceptable latency, effectively making CORBA unusable. The geographical distribution of the network is shown in Fig 2.

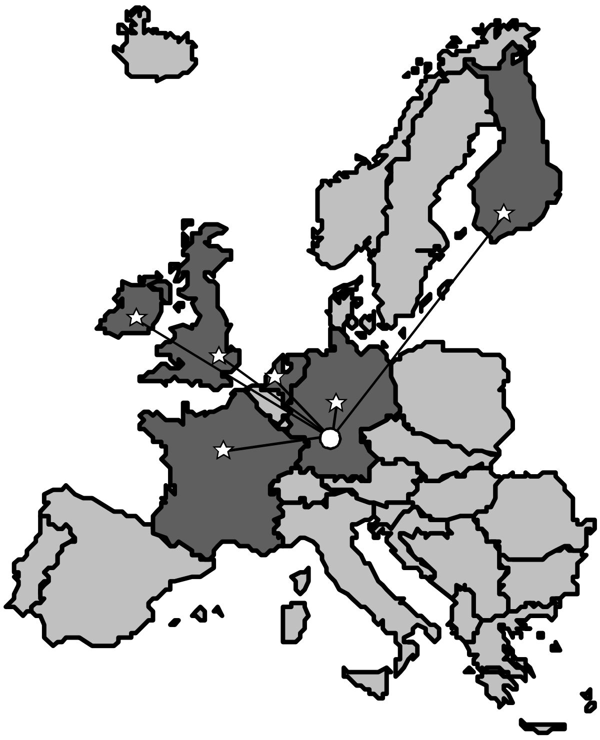

Fig 2 Geographical distribution of the EURESCOM services platform.

The platform consists of six nodes, one at each partner's site, each comprising a LAN with an ISDN2 router. These are connected in a star configuration via a central ISDN router located at EURESCOM headquarters in Heidelberg, Germany. TCP packets between sites are first routed to

Heidelberg on an ISDN connection and then sent on. This approach was taken to simplify both router and host configuration.

At each site there is freedom to choose any combination of hardware, operating system and CORBA middleware. As a result, both Unix and Microsoft Windows are represented, and a variety of ORBs and versions, including:

- Iona Orbix (C++) and OrbixWeb (Java),
- Inprise Visibroker for C++ and Java,
- Sun NEO (C++),
- Distributed Smalltalk,
- Chorus COOL (C++),
- OmniORB,
- Orbacus.

All the object interactions at the service level use CORBA method invocations. There is also an activity on the project to provide stream support on the platform, based on the OMG AV streams specification which will allow more advanced applications to be demonstrated [2].

Each partner has also developed consumer and combined retailer/third party provider components, using TINA reference point specifications — thus providing a set of interoperable service platforms. Two scenarios, deliberately kept as simple as possible, have driven the implementation work. All partners have participated in the first scenario which is based on ‘Ret’ and forms the main content of section 6. Two partners, BT and DT, have also implemented a second scenario based on the third party reference point. This is described in section 8.

6. Interoperability based on ‘Ret’

The TINA Ret reference point [6] is defined to allow interoperability between a consumer and a retailer. The successful implementation of a subset of this reference point should allow any partner’s client (i.e. acting as a consumer) to access any other partner’s retailer and obtain a service.

It is assumed here that, as the retailer and third party service provider are combined in a single enterprise, no additional reference points are required. Reflecting the separation of concerns in the service architecture, interactions at Ret are separated into two parts, an access part and a usage part, as shown in Fig 3. The access part supports a consumer’s interactions with a retailer which are not related to a particular service. It offers the following capabilities:

- initiation of dialogue between consumer and retailer domains,
- identification of the domains to each other,
- establishment of a secure association between the domains,
- set-up of the default context for the control and management of usage functionality,
- discovery of service offerings,
- listings of access sessions, service sessions and subscribed services,
- initiation of usage between the domains,
- control and management of service sessions (suspend/ resume, join, notify changes, etc).

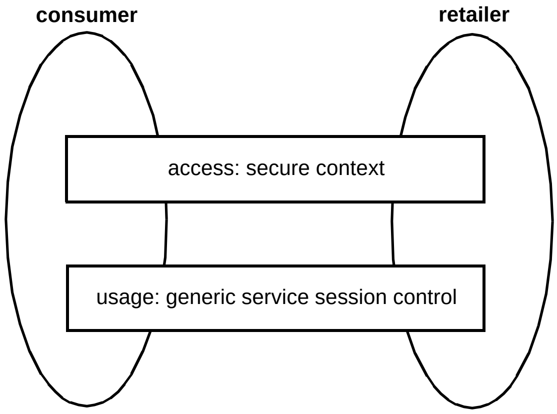

Fig 3 Access and usage parts of the interactions between consumer and retailer across Ret.

The usage part is concerned with generic interactions between various parts of a service instance in providing service to a consumer, using the TINA session model. It is expected that a particular service will make use of the usage part of Ret but will also add its own interfaces to support its own specific requirements.

Conformance to Ret is separate for access and usage parts. The main reason for this is to allow existing services (i.e. non-TINA) to be offered through the access part, as part of a managed portfolio of services. This means that the computational model (depicted in Fig 4) is a simple one.

The provider agent (PA) is the contact point of the retailer within the consumer domain (for the delivery of invitations to join a service session, for example). In these implementations, this component has a user interface and is used by the consumer to control access to services.

The initial agent (IA) is the contact point of the consumer within the retailer domain, and is responsible for authentication and establishment of an access session. It can be regarded as the entry point to the TINA system.

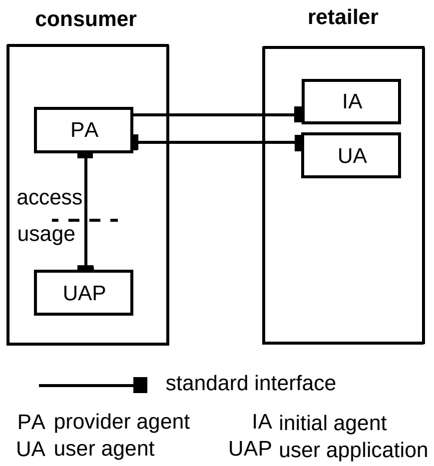

Fig 4 Computational model for consumer/retailer interoperability based on Ret-access.

The user agent (UA) is the contact point of the consumer within the retailer domain once an access session has been established. It represents a single user in the retailer domain and is of the right granularity to deal with the details of individual user accounts, such as preferences, subscriptions and accounting. Multiple simultaneous access sessions can be associated with a UA.

The user application (UAP) is that component of a particular service that executes in the consumer domain. It is the front-end of a distributed application that also offers a user interface to control the service. This interface is required to achieve a basic level of interoperability and is discussed more fully later in this section.

The initial interworking target was for a consumer to access and interact with any retailer for the purpose of using a service supplied by the retailer. This is depicted in Fig 5.

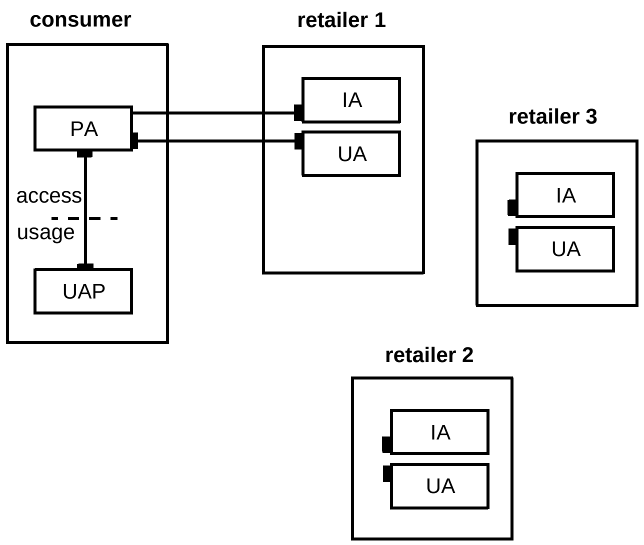

Fig 5 Example of the supported interworking interactions based on Ret.

The PA in the scenario depicted in Fig 5 can be regarded as a ‘browser’ of TINA retailers which is installed in the consumer terminal.

The UAP is part of the specific service. Java bytecode was used, downloaded from the retailer to the consumer at the point of use to avoid having to manage multiple versions of the UAP for different consumer platforms. Other options, such as the pre-installation of platform-specific binaries will be addressed later in the project. The basic sequence for this scenario was:

- start access session,
- select service,
- download consumer domain code,
- use service,
- end access session.

A variant was also considered, which involved the handling of invitations by the consumer:

- start access session,
- accept invitation,
- download consumer domain code,
- use service,
- end access session.

A single retailer was involved in the provision of each service instance. In a multiparty service session, all participants have an access session with the retailer providing the service.

The major functional omissions from the access part of Ret were service session announcements and the ability to suspend and resume sessions.

A subset of Ret was identified to support the above scenarios. In the case without invitations, this subset consists of seven operations in the retailer domain and no consumer domain operations (pure client). In the case with invitations, there are nine operations in the retailer domain and two in the consumer domain.

In addition to the subset of Ret, there was a need to introduce standard interfaces between components executing within the consumer domain (PA and UAP), as indicated in Fig 4. This is not part of the Ret specification, but should be a mandatory element of any statement of compliance whose goal is consumer/retailer interoperability. The reason for this is that the PA is a consumer-domain component which is not specific to any provider. The UAP is part of a particular service and is specific to a particular provider. In order to allow correct initialisation of the UAP by the PA and subsequent interaction, standard interfaces must be available. Simple interfaces supporting the target scenarios have been specified in this project.

Several of the operations specified in Ret make use of property lists (i.e. sequences of ‘string’, ‘any pairs’) to allow general parameter passing. This is important for implementation flexibility, but the details of the properties passed must be agreed if interworking is to be achieved. In the present case, it was found necessary to define a small number of properties associated with the downloading of UAP code into the consumer domain, such as whether it is an applet or application, and a suitable URL.

7. Interoperability issues

Simple services were developed by each participant to demonstrate the test scenarios. These included various games, a shared white board and a multiparty text-chat service. The UAP part of each service was implemented in Java, either as an applet or an application (or both) using a Java ORB (either Visibroker or OrbixWeb).

The use of applets for the UAP adds some complexity. A browser is required to run the applet and this imposes some default security restrictions, which must be overridden to allow execution. This is not possible for all browsers and, where it is, the details are browser-specific. For example, Netscape Communicator 4 also allows applet security to be controlled, but requires browser-specific code in the applet, which limits portability to different browsers.

Although the UAP components were all developed using Java ORBs, the lack of client-side portability means that the appropriate ORB classes had to be available on each consumer machine or downloaded with the UAP bytecode. This introduces a significant delay in downloading. Implementation of Java ORB portability interfaces [7] should greatly improve this aspect of service delivery.

Significant problems were encountered with incompatibilities between certain versions of Java ORBs and some versions of Java. This is perhaps not surprising, as both Java and Java ORBs have been evolving in parallel.

Successful interworking required that diversity of platform was restricted — specifically in the choice of browser and Java version. ORB interoperability generally worked well, although some problems were experienced with:

- marshalling of CORBA 'Any' types containing complex structures,
- narrowing of derived object references, requiring interface repository information from a remote ORB,
- handling of LOCATION_FORWARD messages,
- rebinding to objects after orderly closing of an IIOP connection.

The first two of these issues were avoided in these implementations by using only basic types in 'Anys', and by exporting only objects supporting interfaces explicitly defined in the common IDL specifications. The last two are harder to deal with in implementation, but recent ORB versions have largely resolved these bugs.

Table 1 indicates that, for the basic scenario (no invitations), both consumer-domain ORBs were able to interwork with each of the retailer-domain ORBs across the pan-European network. Some minor issues remain between particular implementations, but there appear to be no major obstacles to progress. At present, only one consumer can accept invitations, and only one service issues them. For this combination (Visibroker for Java/NEO), the second scenario was successfully completed.

Table 1 Summary of interworking without invitation between consumer and retailer domain components implemented using different ORBs.

Consumer	Retailer			
	Orbix	OrbixWeb	NEO	DST
OrbixWeb	✓	✓	✓	✓
Visibroker (Java)	✓	✓	✓	✓

✓: successful completion

The general conclusion from this phase of the work is that the TINA Ret specification contains a sound basis for interoperability between user and provider roles. However, interfaces between consumer domain components and the user application, which originates in the provider domain, must also be agreed.

The Ret reference point is underspecified and requires some additional agreement to ensure interoperability. In addition, CORBA ORBs supporting IIOP can be successfully used to implement a heterogeneous TINA DPE. There are several issues associated with portable executable code, which are still unresolved.

8. Interoperability based on 3Pty

The work described so far discusses a consumer contacting a retailer of choice to access a service provided directly by the retailer. Here a true retailing scenario is considered in which a consumer can access a preferred retailer for the provision of all desired services, including services supplied by a third party. This is shown in Fig 6 where the consumer at the top of the figure accesses services provided by A, B or C.

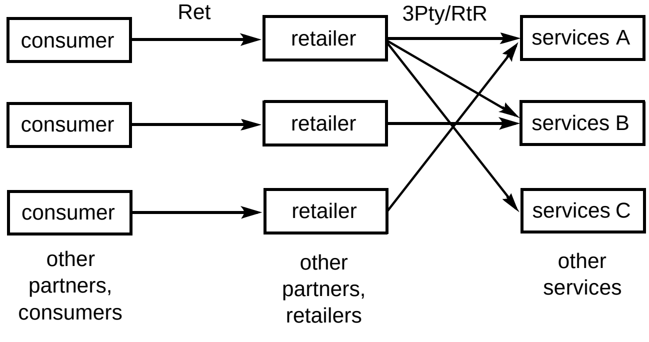

Fig 6 The multiservice provider environment business model.

The interactions between retailer and third party service provider are detailed in the '3Pty' reference point. This has not yet been specified. The service architecture identifies the components envisaged to support interactions at 3Pty. These are shown in Fig 7. The major addition is a new component — the peer agent (peerA), which has characteristics of both a PA and a UA.

The assumption that the access part of Ret can largely be reused is taken as the premise for validation.

The retailer manages the delivery of services from a third party to one of a group of consumers. The third party service provider has no visibility of individual consumers, who, in turn have no direct visibility of the third party service provider. The retailer mediates all access-related interactions and may, but need not, forward requests to the third party service provider. However, usage-related interactions may be directly between the consumer and the third party service provider, bypassing the retailer.

In business terms, it is assumed that contractual agreements exist between consumer and retailer and between retailer and third party service provider, but not between consumer and third party service provider. The interactions between retailer and third party service provider are of the same user-provider type as between consumer and retailer in Ret. The access session also has similar requirements and characteristics. A retailer can establish an access session with a third party either for its own use, independent of any consumer, or on behalf of one or more consumers.

In the first case, the retailer could manage its relationship with the third party service provider, including which services it will retail.

In the second case, the access session may manage a number of service sessions for a number of consumers. Each consumer must have an access session with the retailer. It is up to the retailer to choose how to map from consumer access sessions to its third party access sessions. There are a few issues associated with this aggregation of consumer access sessions that require further study. One implication is that the third party service provider cannot infer the characteristics of the consumer from information associated with the access session. If such information is required, it must be explicitly passed when a consumer starts to use a service.

Another complication is in the use of identifiers to refer to service sessions, invitations, etc. In Ret, these are generated by the retailer and then only used between the consumer and retailer. In this more complex case, multiple third parties may generate duplicate identifiers. This will need to be resolved in a full specification of the reference point, but having identified the problem, a range of identifiers are reserved for each third party service provider in this work.

The handling of invitations is more complicated when multiple retailers are involved in a service instance. It is assumed that the service session is controlled by a single third party. However, the situation now arises where a consumer associated with one retailer wants to invite a consumer who is associated with a different retailer to participate in the service session. This requires that user names be unique, possibly qualified by the retailer name. The form userA@retailerB is used here. The invitation can then be sent to the correct retailer for forwarding to the consumer.

The TINA Working Group responsible for specifying the 3Pty reference point will consider these issues more fully. Having agreed how to proceed, it was possible to reuse the Ret specifications without modification. This has the benefit that changes were only required to the platforms that act in the pure retailer role, as long as invitation propagation is not required. Any of the platforms supporting Ret could then be used as a third party by an enhanced retailer. Parsing of the user name and despatch of an invitation to the correct retailer also requires some modification to platforms acting as third party service providers. The necessary modifications were made to the BT and DT retailer which were then able to offer services from one of the other retailers. These services were

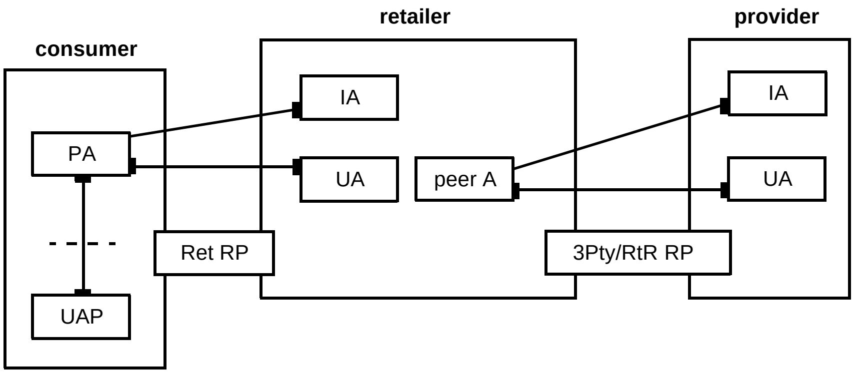

Fig 7 Components within the consumer, retailer and provider domains.

presented to consumers in a uniform way with locally provided services. Propagation of invitations between platforms was also successfully demonstrated between consumers associated with different retailers. Specifically, one retailer was based on Sun NEO on a Solaris platform and the second on ORL OmniORB on a Windows NT platform. No significant problems were encountered beyond the issues already identified. This reinforces the initial view that Ret specifications can be extensively reused to support retailer/third party service provider interactions.

9. Conclusions

The architecture proposed by TINA-C embraces a distributed processing environment and a service management platform that is highly relevant in today's liberalised telecommunications market. The new TINA-C is currently working to achieve the adoption of the TINA-C vision by standards bodies so that not only can stable specifications be produced for topics of particular interest, but also compliant products made available.

This paper has described a collaborative project that has investigated the status of middleware technology and TINA-C specifications by implementing a set of interoperable service platforms. Two scenarios have been successfully demonstrated — an end user accessing a retailer to obtain services directly and an end user accessing a preferred retailer for the provision of all services, including those provided by a third party.

The TINA-C specifications used, although currently incomplete, provided a sound basis for interoperability between consumer, retailer and third party service provider roles. Some behavioural aspects of the objects involved which are not expressed in the IDL specifications must also be agreed. Finally, CORBA ORBs using IIOP can be successfully used to support interworking TINA-C systems. There still remain a number of issues associated with portable executable code, which are not yet fully resolved. It is envisaged that these issues will be resolved in future product releases within this rapidly developing field.

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